



北京大学
PEKING UNIVERSITY

第五章 分子光谱

2. 振动光谱

(一) 双原子分子 振动光谱

分子势能面在平衡位置作泰勒展开

$$V(x) = V(0) + \left(\frac{dV}{dx} \right)_0 x + \frac{1}{2} \left(\frac{d^2V}{dx^2} \right)_0 x^2 + \dots$$

不妨设 $V(0) = 0$

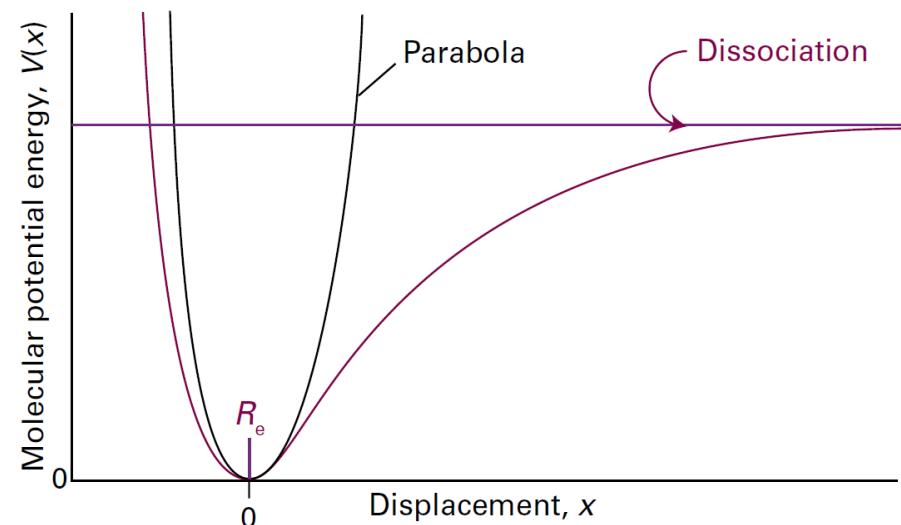
平衡位置一阶项为0

忽略高阶项得 $V(x) \approx \frac{1}{2} \left(\frac{d^2V}{dx^2} \right)_0 x^2$

势能近似为抛物线

$$V(x) = \frac{1}{2} k_f x^2 \quad x = R - R_e$$

$$k_f = \left(\frac{d^2V}{dx^2} \right)_0 \quad k_f : \text{力常数}$$



简谐振子模型

$$-\frac{\hbar^2}{2m_{\text{eff}}} \frac{d^2\psi}{dx^2} + \frac{1}{2} k_f x^2 \psi = E\psi$$

$$m_{\text{eff}} = \frac{m_1 m_2}{m_1 + m_2}$$

振动是相对运动，采用有效质量

复习谐振子能级

$$E_v = \left(v + \frac{1}{2}\right) \hbar \omega \quad \omega = \left(\frac{k_f}{m_{\text{eff}}}\right)^{1/2} \quad v = 0, 1, 2, \dots$$

改写成波数 (vibrational term)

$$E_v = hc\tilde{G}(v) \quad \tilde{G}(v) = \left(v + \frac{1}{2}\right) \tilde{\nu} \quad \tilde{\nu} = \frac{1}{2\pi c} \left(\frac{k_f}{m_{\text{eff}}}\right)^{1/2}$$

室温下一般处于振动基态

$$kT/hc \approx 200 \text{ cm}^{-1}$$

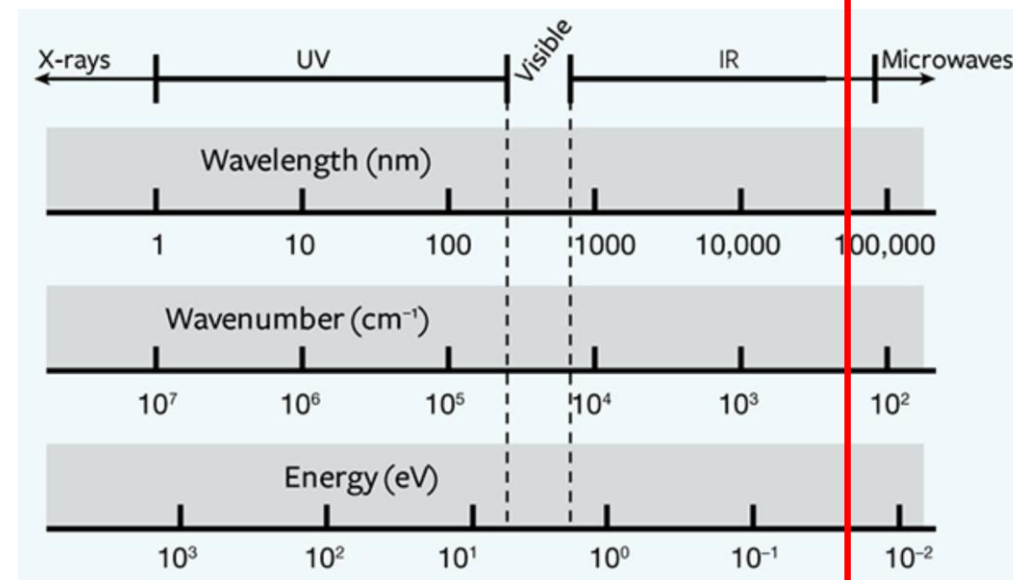
红外光谱

红外活性的跃迁对应非零跃迁偶极矩

- 振动导致偶极矩发生改变
- $\Delta v = \pm 1$

振动跃迁波数

$$\Delta\tilde{G}_{v+\frac{1}{2}} = \tilde{G}(v+1) - \tilde{G}(v) = \tilde{\nu}$$



跃迁选律1: 跃迁偶极矩不为零

跃迁选律2: 跃迁前后偶极矩发生改变

$$\mu_f = \langle \Psi_0 + \lambda \Psi_f | \hat{\mu} | \Psi_0 + \lambda \Psi_f \rangle \quad \text{假定跃迁矩阵元是实数}$$

$$= \langle \Psi_0 | \hat{\mu} | \Psi_0 \rangle + 2\lambda \langle \Psi_f | \hat{\mu} | \Psi_0 \rangle + \lambda^2 \langle \Psi_f | \hat{\mu} | \Psi_f \rangle$$

$$\mu_f - \mu_0 \approx 2\lambda \langle \Psi_f | \hat{\mu} | \Psi_0 \rangle + \mathcal{O}(\lambda^2)$$

一阶微扰近似下, 偶极矩改变量正比于跃迁偶极矩

该结论适用于振动、转动等各种吸收跃迁

$$\alpha_f - \alpha_0 \approx 2\lambda \langle \Psi_f | \hat{\alpha} | \Psi_0 \rangle + \mathcal{O}(\lambda^2) \quad \text{极化率也有类似结论}$$

谐振子近似下，所有跃迁频率相同，只有一条谱线

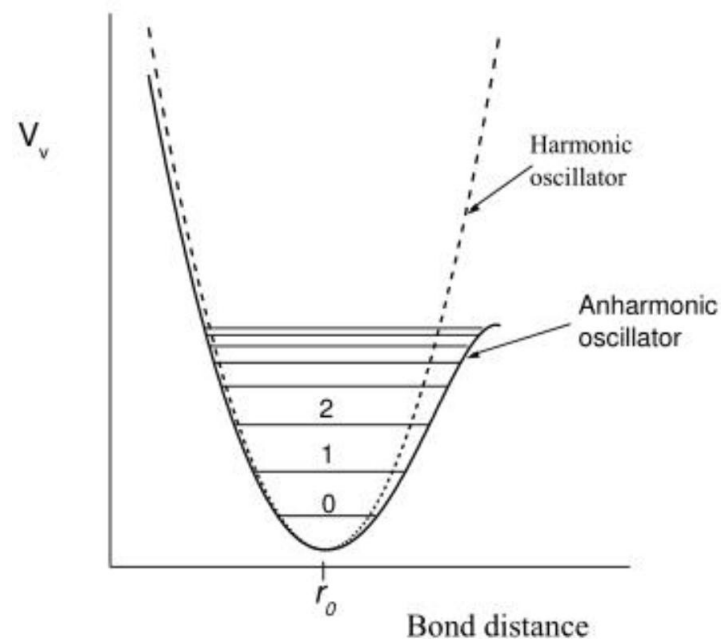
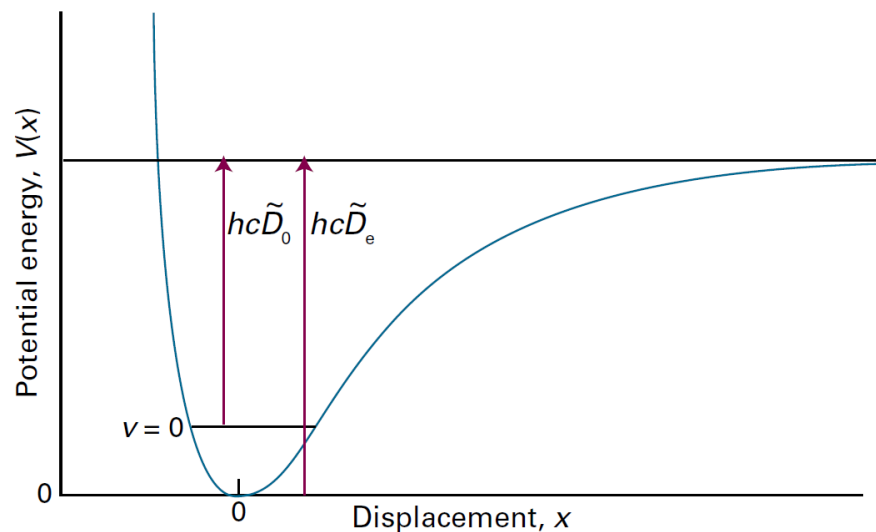
其中， $1 \leftarrow 0$ 是fundamental transition

实际情况，多条谱线

非谐性 (anharmonicity)

势能偏离二次势

等距能级 $\rightarrow$ 非等距能级



真实势能面在无穷远处不发散

键解离能: $hc\tilde{D}_0$

势井深度: $hc\tilde{D}_e$

} 两者相差振动零点能

摩尔斯势 (Morse potential)
考虑非谐性对简谐势的改进

$$V(x) = hc\tilde{D}_e \{1 - e^{-ax}\}^2 \quad a = \left(\frac{m_{\text{eff}} \omega^2}{2hc\tilde{D}_e} \right)^{1/2}$$

平衡位置展开截断到二阶等
价于简谐势

能级公式

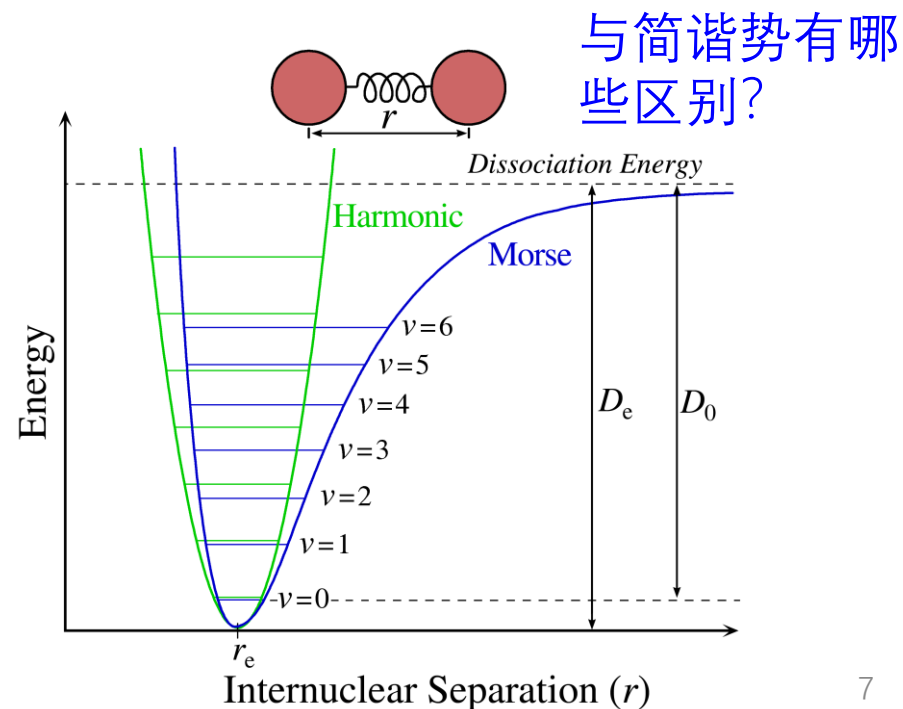
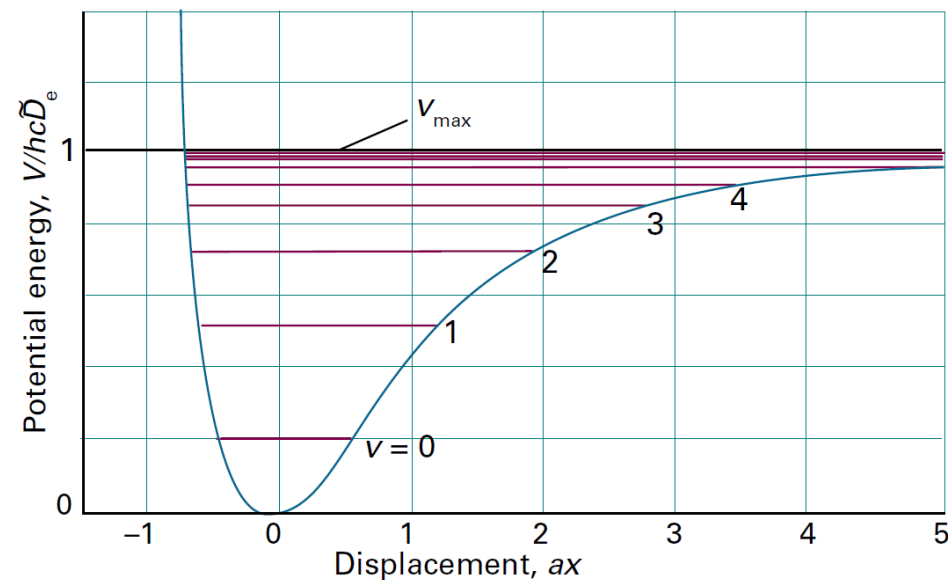
$$\tilde{G}(v) = \left(v + \frac{1}{2}\right) \tilde{\nu} - \left(v + \frac{1}{2}\right)^2 x_e \tilde{\nu}$$

$$v = 0, 1, 2, \dots, v_{\text{max}}$$

$$x_e = \frac{a^2 \hbar}{2m_{\text{eff}} \omega} = \frac{\tilde{\nu}}{4\tilde{D}_e}$$

有限个振动态

非谐性常数



更一般的经验公式

$$\tilde{G}(\nu) = (\nu + \frac{1}{2})\tilde{\nu} - (\nu + \frac{1}{2})^2 x_e \tilde{\nu} + (\nu + \frac{1}{2})^3 y_e \tilde{\nu} + \dots$$

$\tilde{G}$ 对 $\nu + \frac{1}{2}$ 泰勒展开, x_e, y_e 通过实验拟合

1 $\leftarrow$ 0: 基础跃迁

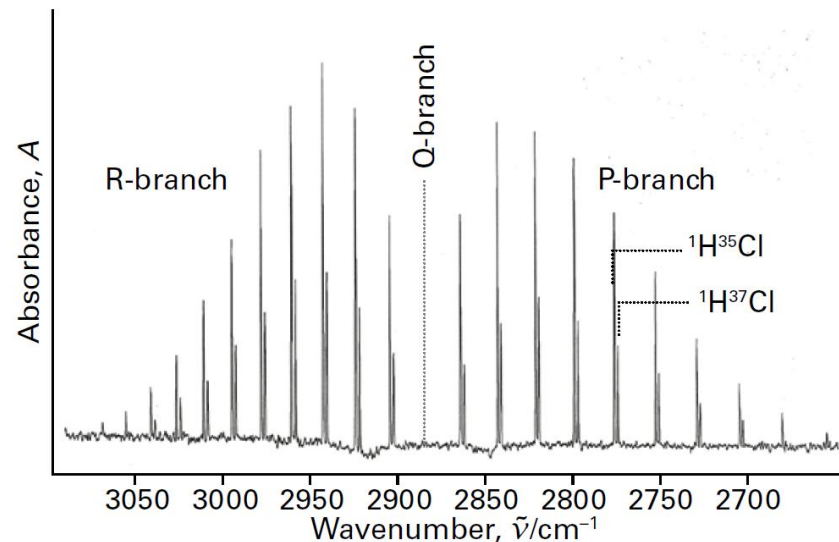
2 $\leftarrow$ 0: 第一泛音 (first overtone)

3 $\leftarrow$ 0: 第二泛音 (second overtone)

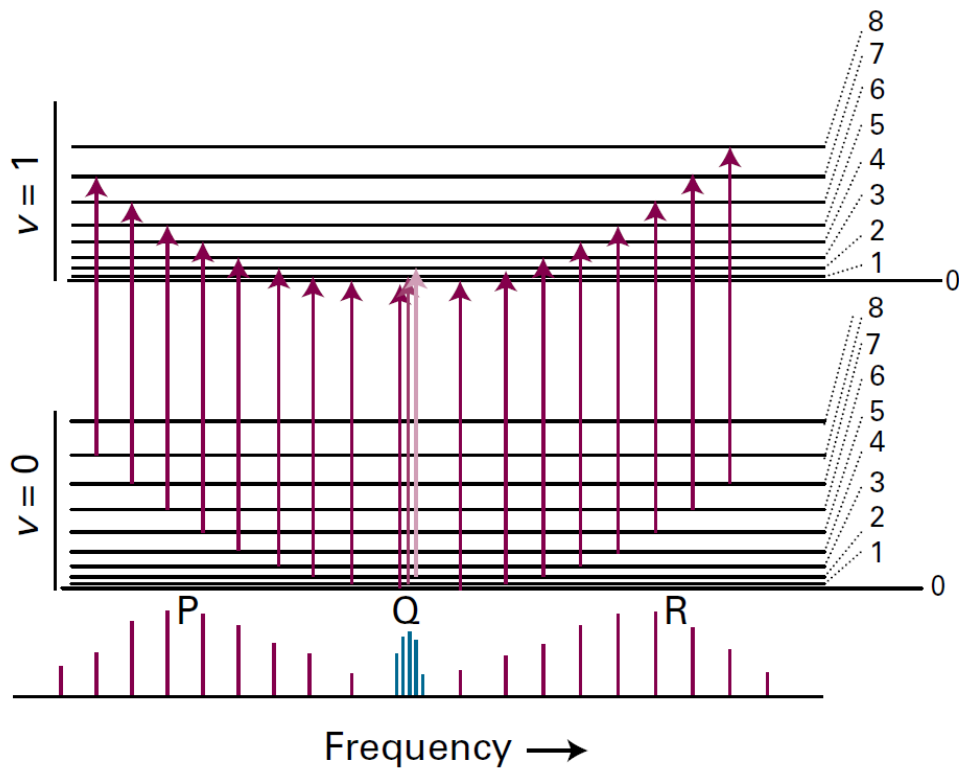
简谐近似下没有泛音, 实际情况泛音存在, 但比较弱

实际红外谱叠加转动, 形成谱带

振动-转动谱



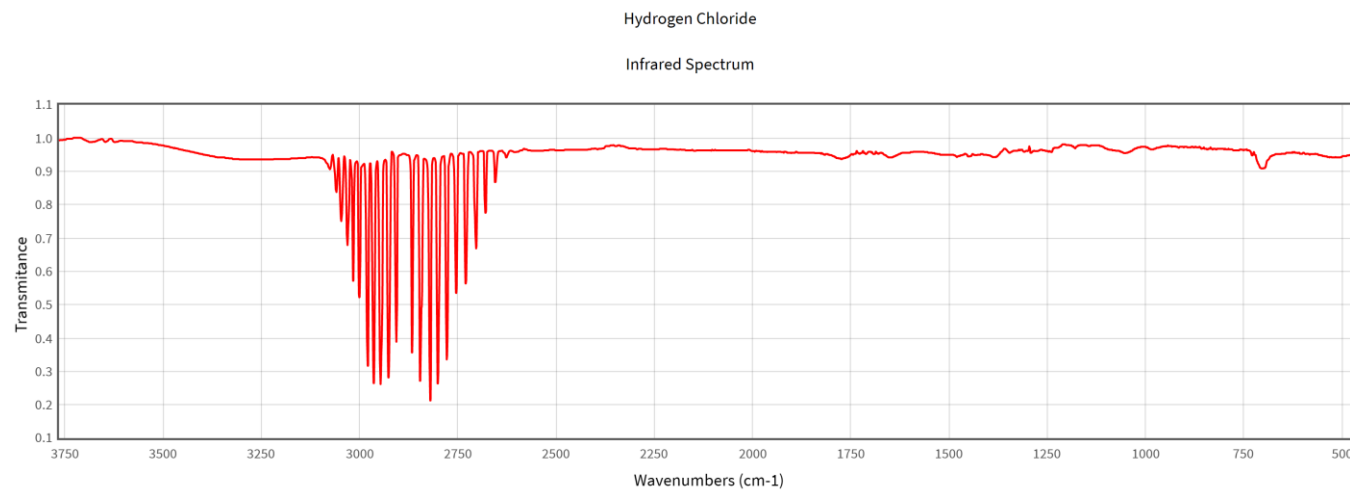
高清分辨的谱图, 形成P, Q, R三个分支



P 支: $\Delta J = -1$

Q 支: $\Delta J = 0$

R 支: $\Delta J = 1$

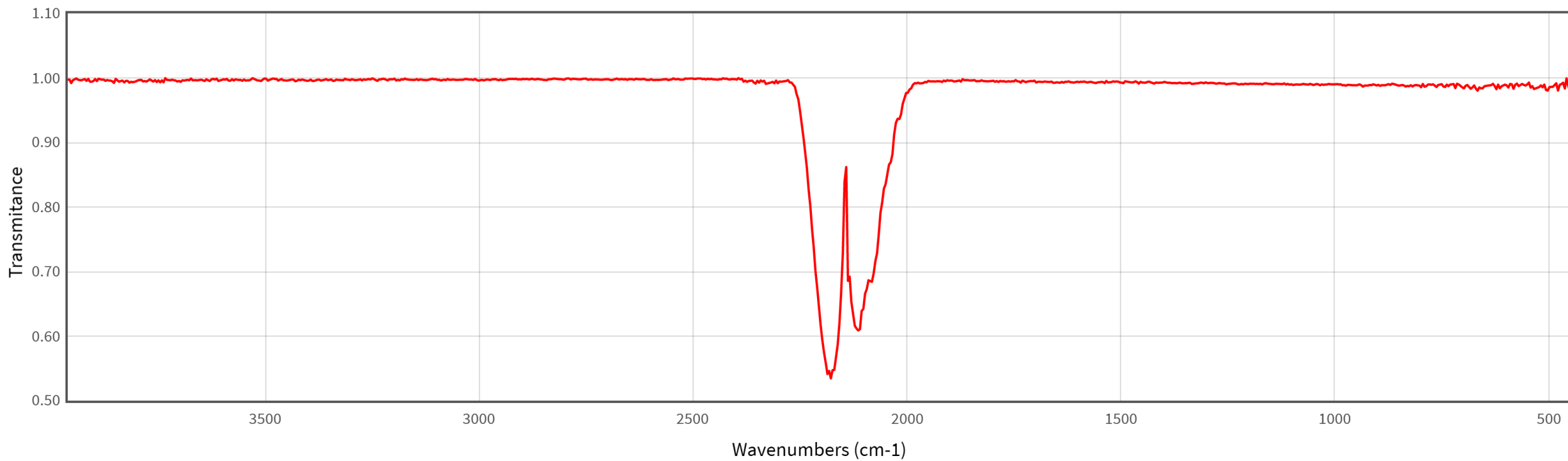


<https://webbook.nist.gov/chemistry/>

	$\tilde{\nu}/\text{cm}^{-1}$	R_e/pm	$\tilde{B}/\text{cm}^{-1}$	$k_f/(\text{N m}^{-1})$	$\tilde{D}_0/(10^4 \text{ cm}^{-1})$
$^1\text{H}_2$	4400	74	60.86	575	3.61
$^1\text{H}^{35}\text{Cl}$	2991	127	10.59	516	3.58
$^1\text{H}^{127}\text{I}$	2308	161	6.51	314	2.46
$^{35}\text{Cl}_2$	560	199	0.244	323	2.00

Carbon monoxide

Infrared Spectrum

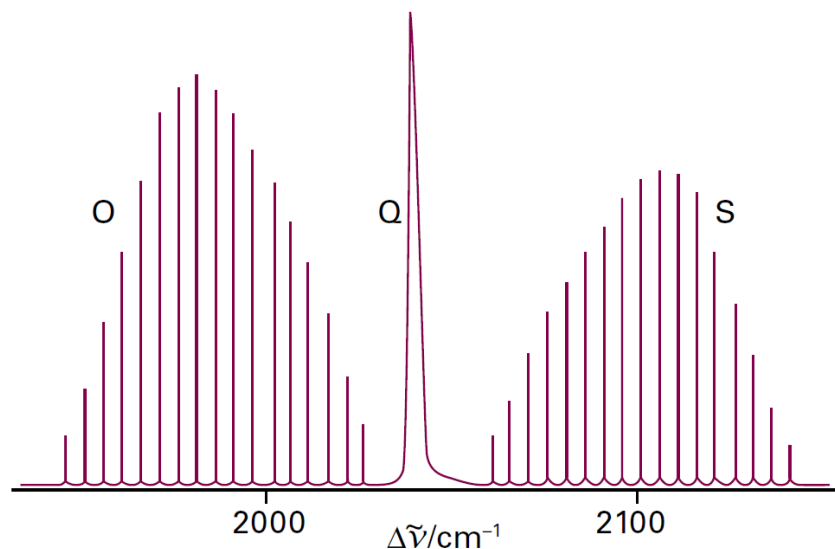


振动拉曼谱

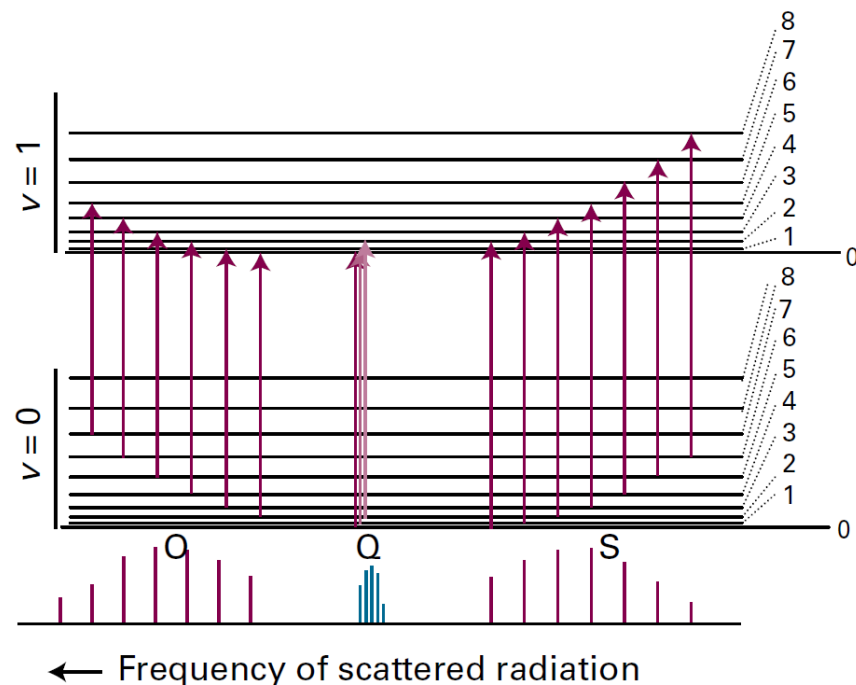
选律:

- 振动导致极化率发生改变
- $\Delta v = \pm 1$, 反Stokes跃迁弱

振动-转动光谱图,
形成O, Q, S三个分支



CO拉曼谱, 中心峰有展宽



O 支: $\Delta J = -2$

Q支: $\Delta J = 0$

S 支: $\Delta J = 2$

O P Q R S

(二) 多原子分子 振动光谱

自由度

N个原子有 $3N$ 个坐标（自由度）

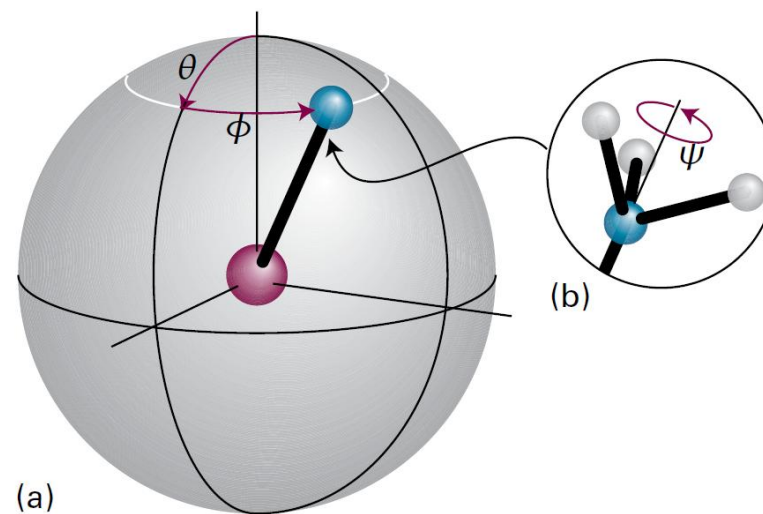
总自由度 = 平动 + 振动 + 转动

平动自由度: 3

转动自由度 $\left\{ \begin{array}{l} \text{线性分子: } 2 \\ \text{非线性分子: } 3 \end{array} \right.$

振动自由度 $\left\{ \begin{array}{l} \text{线性分子: } 3N-5 \\ \text{非线性分子: } 3N-6 \end{array} \right.$

又叫振动模式
(vibrational modes)



CO_2 : 4个振动模式

H_2O : 3个振动模式

甲苯: 15个原子, 39个振动模式

正则模式 (normal mode)

以水分子为例，势能面可以视为三个键长的函数

$$E = E(r_1, r_2, r_3)$$

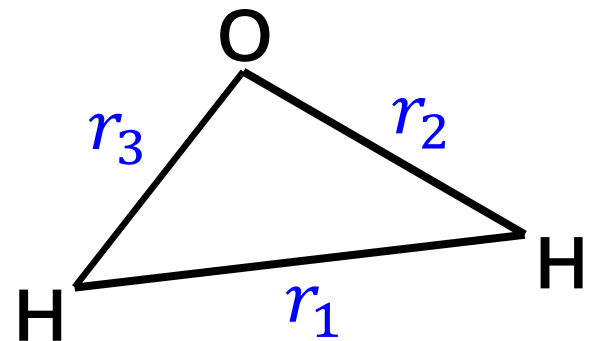
平衡键长为 $r_1^0 = 124.5 \text{ pm}$ $r_2^0 = r_3^0 = 95.84 \text{ pm}$

能量在平衡键长作三维泰勒展开

$$E(r_1, r_2, r_3) = E_0 + \sum_k \frac{\partial E}{\partial r_k} \Big|_0 (r_k - r_k^0) + \frac{1}{2} \sum_{jk} \frac{\partial^2 E}{\partial r_j \partial r_k} \Big|_0 (r_j - r_j^0)(r_k - r_k^0) + \dots$$

不妨令 $E_0 = E(r_1^0, r_2^0, r_3^0) = 0$

平衡键长意味着 $\frac{\partial E}{\partial r_k} \Big|_0 = \frac{\partial E}{\partial r_k} \Big|_{(r_1, r_2, r_3) = (r_1^0, r_2^0, r_3^0)} = 0$



能量截断到二次项，得

$$E(r_1, r_2, r_3) = \frac{1}{2} \sum_{jk} \left. \frac{\partial^2 E}{\partial r_j \partial r_k} \right|_0 (r_j - r_j^0)(r_k - r_k^0)$$

用位移改写

$$E(q_1, q_2, q_3) = \frac{1}{2} \sum_{jk} H_{jk} q_j q_k$$

$$q_k = r_k - r_k^0$$

进一步改写成向量二次型

$$H_{jk} = \left. \frac{\partial^2 E}{\partial r_j \partial r_k} \right|_0 = \left. \frac{\partial^2 E}{\partial q_j \partial q_k} \right|_0$$

$$E(\mathbf{q}) = \mathbf{q}^T \mathbf{H} \mathbf{q}$$

$$\mathbf{q} = (q_1, q_2, q_3)$$

位移排成向量

$$\mathbf{H} = \begin{bmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & H_{33} \end{bmatrix}$$

海瑟矩阵



$$\mathbf{E} = \begin{bmatrix} \mathcal{E}_1 & 0 & 0 \\ 0 & \mathcal{E}_2 & 0 \\ 0 & 0 & \mathcal{E}_3 \end{bmatrix}$$

正则(简正)模式

总结正则模式

$$E(\mathbf{q}) = \mathbf{q}^T \mathbf{H} \mathbf{q} = \mathbf{Q}^T \boldsymbol{\varepsilon} \mathbf{Q} = \sum_{j=1}^{N_{\text{vib}}} \epsilon_j Q_j^2$$

$$\mathbf{H} = \mathbf{U}^T \boldsymbol{\varepsilon} \mathbf{U} \quad \mathbf{Q} = \mathbf{U} \mathbf{q} \quad \mathbf{Q} \text{ 是正则模式坐标, 是非正则模式的酉变换}$$

简谐振动哈密顿量

$$\hat{H}_{\text{vib}} = \sum_{j=1}^{N_{\text{vib}}} \left(-\frac{\hbar^2}{2\mu_j} \frac{\partial^2}{\partial Q_j^2} + \frac{1}{2} \epsilon_j Q_j^2 \right) \quad \text{各振动模式相互独立, } \mu_j \text{ 是振动模式 } j \text{ 对应的有效质量}$$

$$\hat{H}_{\text{nuclei}} = -\sum_{j=1}^N \frac{\hbar^2}{2M_j} \nabla_{\mathbf{R}_j}^2 + E_{\text{BO}}(\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_N)$$

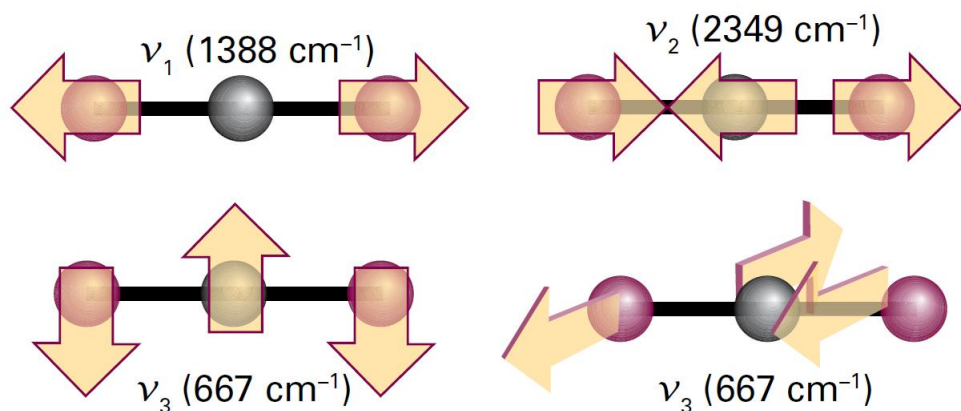
从 $\hat{H}_{\text{nuclei}}$ 到 $\hat{H}_{\text{vib}}$, 经过了 **近似的** 变量分离步骤
(分离质心平动、整体转动, 进而推得 μ_j)

简正模式一般是所有原子参与的协同运动, 但质心不动

简正模式的能量

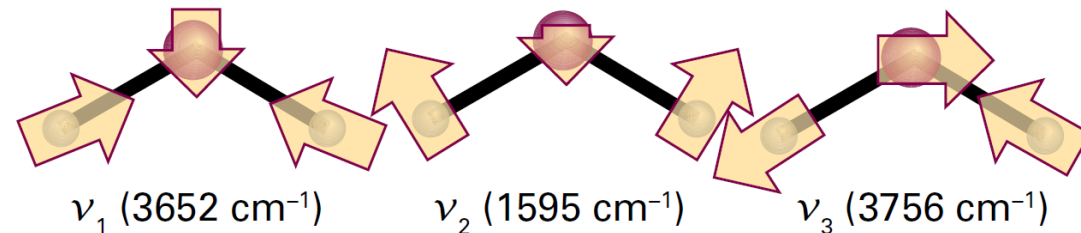
$$\tilde{G}_q(\nu) = (\nu_q + \frac{1}{2})\tilde{\nu}_q \quad \nu_q = 0, 1, 2, \dots \quad \tilde{\nu}_q = \frac{1}{2\pi c} \left(\frac{k_{f,q}}{m_q} \right)^{1/2}$$

正则模式坐标 Q
可以用键长、键
角重新表示



CO₂简正模式

对称伸缩, 反对称伸缩(stretch)
2个简并弯折(bending)



H₂O简正模式

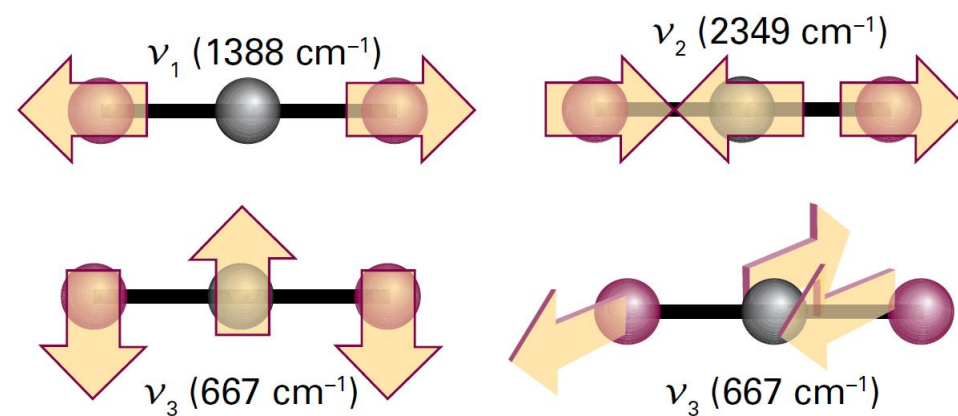
兼有伸缩和弯折 (混合模式)

伸缩: 只改变键长, 键角不变

弯折: 只改变键角, 键长不变

红外吸收选律:

- 某一简正模式有吸收则跃迁一定伴随**偶极矩**变化
- $\Delta v_q = \pm 1$ (简正近似下)



对称伸缩无红外活性, 其他模式有红外活性

fundamental transition

以水分子为例

$(1, 0, 0) \leftarrow (0, 0, 0)$

$(0, 1, 0) \leftarrow (0, 0, 0)$

$(0, 0, 1) \leftarrow (0, 0, 0)$

考虑到非谐性, 会出现泛音, 如

$(2, 0, 0) \leftarrow (0, 0, 0)$ $(1, 1, 0) \leftarrow (0, 0, 0)$

一般多原子分子气相振动谱很难分析
(许多简正模式, 许多泛音, 转动精细结构)

从基态吸收光发生跃迁

跃迁偶极矩 $\mu_{f0} = \langle \Psi_f | \hat{\mu} | \Psi_0 \rangle$ $\hat{\mu} = \sum_k Z_k \mathbf{R}_k - \sum_j \mathbf{r}_j$ 偶极算符

基态波函数 $\Psi_0(\underline{r}, \underline{R}) = \chi_0(\underline{R}) \Phi_{\underline{R}}^0(\underline{r})$ $\Psi_f(\underline{r}, \underline{R}) = \chi_f(\underline{R}) \Phi_{\underline{R}}^0(\underline{r})$ 终态波函数

$$\mu_{f0}^n = \langle \Psi_f | \sum_k Z_k \mathbf{R}_k | \Psi_0 \rangle_{\underline{r}\underline{R}} = \langle \chi_f(\underline{R}) \Phi_{\underline{R}}^0(\underline{r}) | \sum_k Z_k \mathbf{R}_k | \chi_0(\underline{R}) \Phi_{\underline{R}}^0(\underline{r}) \rangle_{\underline{r}\underline{R}}$$

$$= \langle \chi_f(\underline{R}) | \sum_k Z_k \mathbf{R}_k | \chi_0(\underline{R}) \rangle_{\underline{R}}$$

电子波函数归一

$$\mu_{f0}^e = \langle \Psi_f | - \sum_j \mathbf{r}_j | \Psi_0 \rangle_{\underline{r}} = \langle \chi_f(\underline{R}) \Phi_{\underline{R}}^0(\underline{r}) | - \sum_j \mathbf{r}_j | \chi_0(\underline{R}) \Phi_{\underline{R}}^0(\underline{r}) \rangle_{\underline{r}\underline{R}}$$

$$= \langle \chi_f(\underline{R}) | \mu_e(\underline{R}) | \chi_0(\underline{R}) \rangle_{\underline{R}} \quad \mu_e(\underline{R}) = \langle \Phi_{\underline{R}}^0(\underline{r}) | - \sum_j \mathbf{r}_j | \Phi_{\underline{R}}^0(\underline{r}) \rangle_{\underline{r}}$$

$$\mu_{f0} = \mu_{f0}^n + \mu_{f0}^e = \langle \chi_f(\underline{R}) | \mu(\underline{R}) | \chi_0(\underline{R}) \rangle_{\underline{R}} \quad \mu(\underline{R}) = \sum_k Z_k \mathbf{R}_k + \mu_e(\underline{R})$$

$$\mu_{f0} = \mu_{f0}^n + \mu_{f0}^e = \langle \chi_f(\underline{\underline{\mathbf{R}}}) | \underline{\underline{\mu}}(\underline{\underline{\mathbf{R}}}) | \chi_0(\underline{\underline{\mathbf{R}}}) \rangle_{\underline{\underline{\mathbf{R}}}} \quad \underline{\underline{\mu}}(\underline{\underline{\mathbf{R}}}) = \sum_k Z_k \mathbf{R}_k + \underline{\underline{\mu}}_e(\underline{\underline{\mathbf{R}}})$$

分离平动、振动、转动自由度

$$\chi(\underline{\underline{\mathbf{R}}}) \approx \chi_{\text{trans}}(\mathbf{R}_{\text{cm}}) \chi_{\text{rot}}(\theta_{\text{rot}}^1, \theta_{\text{rot}}^2, \theta_{\text{rot}}^3) \chi_{\text{vib}}(Q_{\text{vib}}^1, Q_{\text{vib}}^2, \dots)$$

平动不改变跃迁偶极矩

$$\mu_{f0} = \langle \chi_{\text{rot}}^f(\boldsymbol{\theta}_{\text{rot}}) \chi_{\text{vib}}^f(\mathbf{Q}_{\text{vib}}) | \underline{\underline{\mu}}(\boldsymbol{\theta}_{\text{rot}}, \mathbf{Q}_{\text{vib}}) | \chi_{\text{rot}}^0(\boldsymbol{\theta}_{\text{rot}}) \chi_{\text{vib}}^0(\mathbf{Q}_{\text{vib}}) \rangle_{\boldsymbol{\theta}_{\text{rot}} \mathbf{Q}_{\text{vib}}}$$

先对转动自由度积分 $\tilde{\mu}(\mathbf{Q}_{\text{vib}}) = \langle \chi_{\text{rot}}^f(\boldsymbol{\theta}_{\text{rot}}) | \underline{\underline{\mu}}(\boldsymbol{\theta}_{\text{rot}}, \mathbf{Q}_{\text{vib}}) | \chi_{\text{rot}}^0(\boldsymbol{\theta}_{\text{rot}}) \rangle_{\boldsymbol{\theta}_{\text{rot}}}$

$$\mu_{f0} = \langle \chi_{\text{vib}}^f(\mathbf{Q}_{\text{vib}}) | \tilde{\mu}(\mathbf{Q}_{\text{vib}}) | \chi_{\text{vib}}^0(\mathbf{Q}_{\text{vib}}) \rangle_{\mathbf{Q}_{\text{vib}}}$$

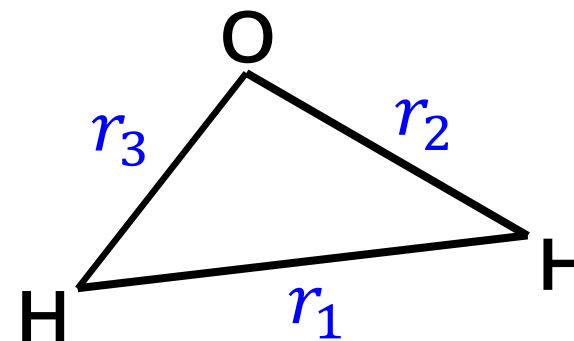
$$\chi_{\text{vib}}^f(\mathbf{Q}_{\text{vib}}) = G_{f_1}(Q_1; \omega_1, \mu_1) G_{f_2}(Q_2; \omega_2, \mu_2) \cdots \quad \chi_{\text{vib}}^0(\mathbf{Q}_{\text{vib}}) = G_0(Q_1; \omega_1, \mu_1) G_0(Q_2; \omega_2, \mu_2) \cdots$$

$$\mu_{f0} = \langle \chi_{\text{vib}}^f(Q_{\text{vib}}) | \tilde{\mu}(Q_{\text{vib}}) | \chi_{\text{vib}}^0(Q_{\text{vib}}) \rangle_{Q_{\text{vib}}}$$

$$\chi_{\text{vib}}^f(Q_{\text{vib}}) = G_{f_1}(Q_1; \omega_1, \mu_1) G_{f_2}(Q_2; \omega_2, \mu_2) \cdots \quad \chi_{\text{vib}}^0(Q_{\text{vib}}) = G_0(Q_1; \omega_1, \mu_1) G_0(Q_2; \omega_2, \mu_2) \cdots$$

$$G_0(Q_1; \omega_1, \mu_1) = \left(\frac{\mu_1 \omega_1}{\pi} \right)^{1/4} e^{-\frac{1}{2} \mu_1 \omega_1 Q_1^2}$$

$$Q_1 = \sum_{k=1}^3 U_{1k} q_k = \sum_{k=1}^3 U_{1k} (r_k - r_k^0)$$



Q_1 是正则模式坐标，以水分子为例，是3个
平衡键长偏移量的线性组合

假定振动模式不简并

以 χ_{vib}^0 作基表示分子点群，为 A_1 不可约表示（任意对称操作下 χ_{vib}^0 均不变）

以 χ_{vib}^f 作基表示分子点群，为某 Γ 不可约表示

$$A_1 \otimes \Gamma = \Gamma$$

以 $\tilde{\mu}_x(Q_{\text{vib}}), \tilde{\mu}_y(Q_{\text{vib}}), \tilde{\mu}_z(Q_{\text{vib}})$ 为基表示分子点群, 与以 x, y, z 为基相同

以某转动操作 $\mathbf{R}$ 为例, 以 (x, y, z) 为基, 表示矩阵为

$$\mathbf{R} : (x, y, z) \rightarrow (x, y, z) \begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix}$$

则以偶极矩三分量为基, 表示矩阵也为

$$\mathbf{R} : (\tilde{\mu}_x, \tilde{\mu}_y, \tilde{\mu}_z) \rightarrow (\tilde{\mu}_x, \tilde{\mu}_y, \tilde{\mu}_z) \begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix}$$

$$\mu_{f0} = \langle \chi_{\text{vib}}^f(\mathbf{Q}_{\text{vib}}) | \tilde{\boldsymbol{\mu}}(\mathbf{Q}_{\text{vib}}) | \chi_{\text{vib}}^0(\mathbf{Q}_{\text{vib}}) \rangle_{\mathbf{Q}_{\text{vib}}}$$

(1) 以 $\tilde{\mu}_x(Q_{\text{vib}}), \tilde{\mu}_y(Q_{\text{vib}}), \tilde{\mu}_z(Q_{\text{vib}})$ 为基进行表示后约化

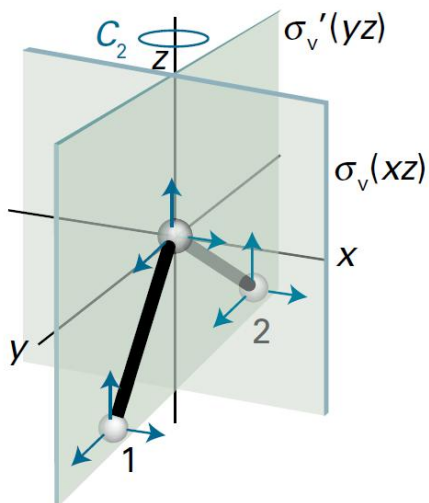
(2) 以 x, y, z 为基进行表示后约化

(1)与(2)得到相同的不可约表示, 特征标也相同

判断 μ_{f0} 是否非零, 只需判断

Γ 不可约表示是否与 x, y, z 对应的不可约表示相同

水分子的振动



1. 以每个原子的笛卡尔坐标为基，得到分子点群的表示

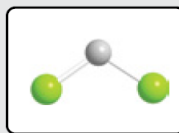
2. 计算该可约表示 Γ 的特征标

$$\chi(E) = 9 \quad \chi(C_2) = -1 \quad \text{只有O原子有贡献: } 1-1-1=-1$$

$$\chi(\sigma_v) = 1 \quad \text{只有O原子有贡献: } 1+1-1=1$$

$$\chi(\sigma'_v) = 3 \quad \text{每个原子都有贡献: } 3 \times (1+1-1)=3$$

$C_{2v}, 2mm$	E	C_2	σ_v	σ'_v	$h = 4$	
A_1	1	1	1	1	z, z^2, x^2, y^2	
A_2	1	1	-1	-1	xy	R_z
B_1	1	-1	1	-1	x, zx	R_y
B_2	1	-1	-1	1	y, yz	R_x



3. 将 Γ 进行约化

$$\Gamma = 3A_1 + A_2 + 2B_1 + 3B_2$$

4. 剔除平动与转动，得到振动模式

$$T = A_1 + B_1 + B_2 \quad \longrightarrow \quad V = 2A_1 + B_2$$

$$R = A_2 + B_1 + B_2$$

都有红外活性

振动光谱红外活性 $\longrightarrow \mu_{f0} = \langle \chi_{\text{vib}}^f(Q_{\text{vib}}) | \tilde{\mu}(Q_{\text{vib}}) | \chi_{\text{vib}}^0(Q_{\text{vib}}) \rangle_{Q_{\text{vib}}}$

振动拉曼活性 $\longrightarrow \alpha_{f0} = \langle \chi_{\text{vib}}^f(Q_{\text{vib}}) | \tilde{\alpha}(Q_{\text{vib}}) | \chi_{\text{vib}}^0(Q_{\text{vib}}) \rangle_{Q_{\text{vib}}}$

以极化率张量为基表示分子点群，与以如下矩阵元素为基相同

$$\alpha = \begin{bmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{bmatrix} \longleftrightarrow \begin{bmatrix} x^2 & xy & xz \\ xy & y^2 & yz \\ xz & yz & z^2 \end{bmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} (x, y, z)$$

判断 α_{f0} 是否非零，只需判断 Γ 不可约表示是否与

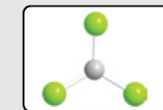
$x^2, y^2, z^2, xy, yz, zx$ 对应的不可约表示相同

有一个相同，跃迁矩阵元即非零，即有拉曼活性

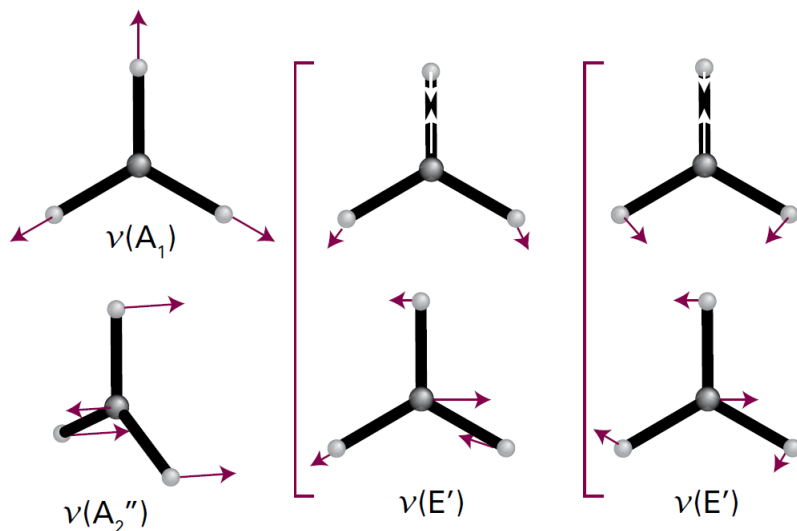
水分子三个振动模式都有拉曼活性

BF₃的振动

$D_{3h}, \bar{6}2m$	E	σ_h	$2C_3$	$2S_3$	$3C_2$	$3\sigma_v$	$h = 12$	
A'_1	1	1	1	1	1	1	$z^2, x^2 + y^2$	
A'_2	1	1	1	1	-1	-1		R_z
A''_1	1	-1	1	-1	1	-1		
A''_2	1	-1	1	-1	-1	1	z	
E'	2	2	-1	-1	0	0	$(x, y), (xy, x^2 - y^2)$	
E''	2	-2	-1	1	0	0	(yz, zx)	(R_x, R_y)



$$\Gamma = A'_1 + A'_2 + 3E' + 2A''_2 + E''$$



$$T = A''_2 + E'$$

$$R = A'_2 + E''$$



$$V = A'_1 + 2E' + A''_2$$



简并振动模式

思考：哪些模式有
红外活性？哪些有
拉曼活性？

回想：水分子
没有简并模
式，跃迁从
 $0 \rightarrow 1$

红外谱的正确使用方法

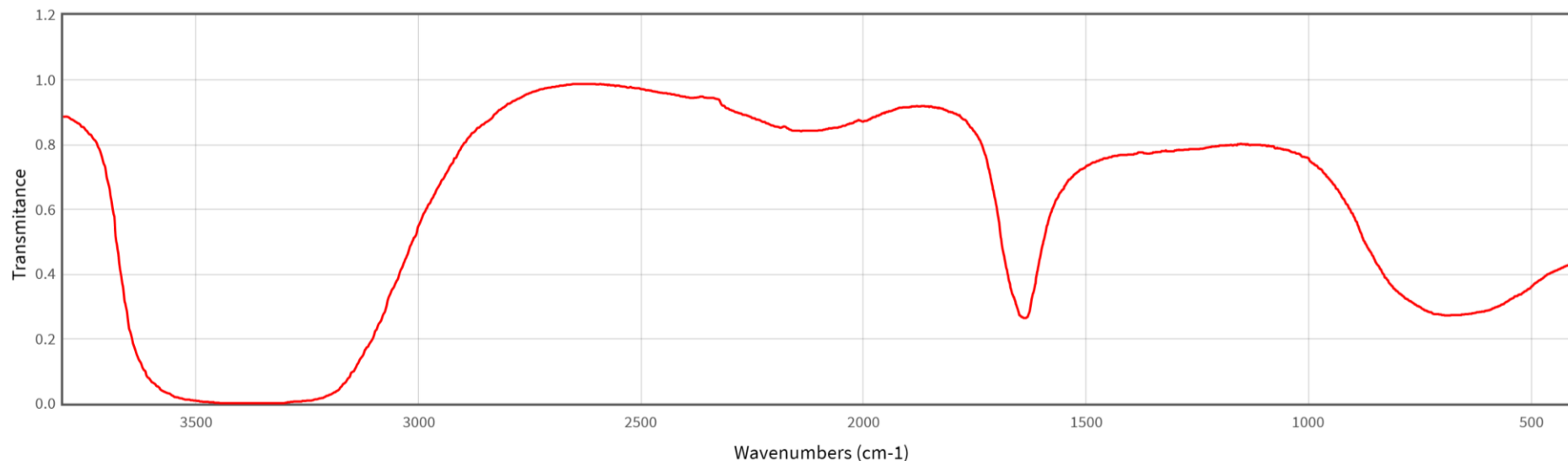
- 凝聚态（液相、固相、溶液）读谱

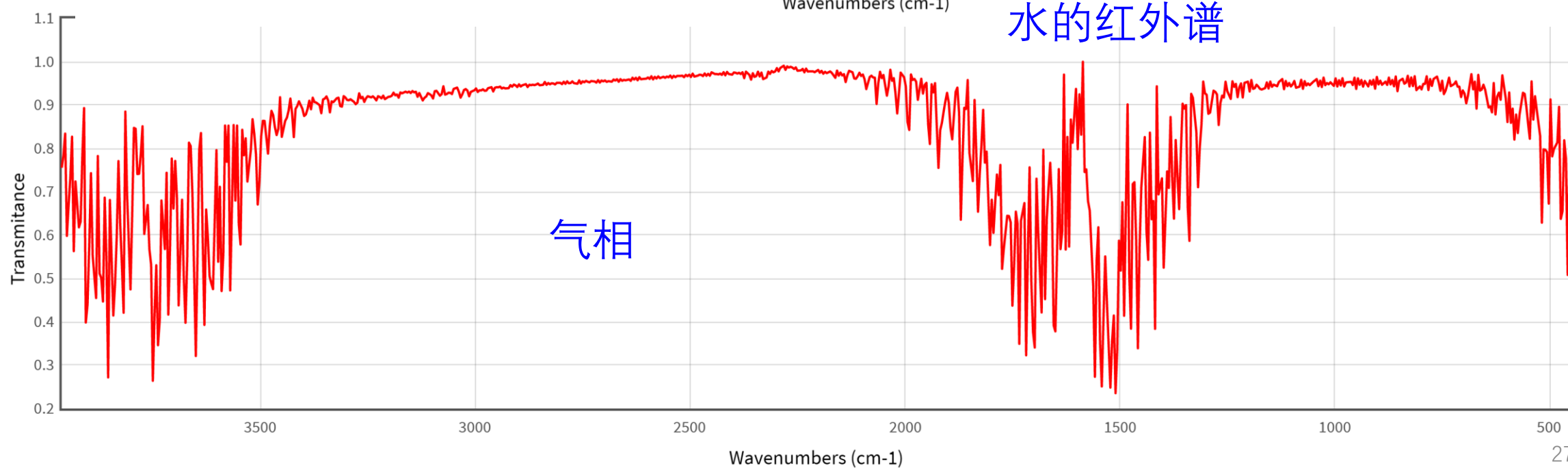
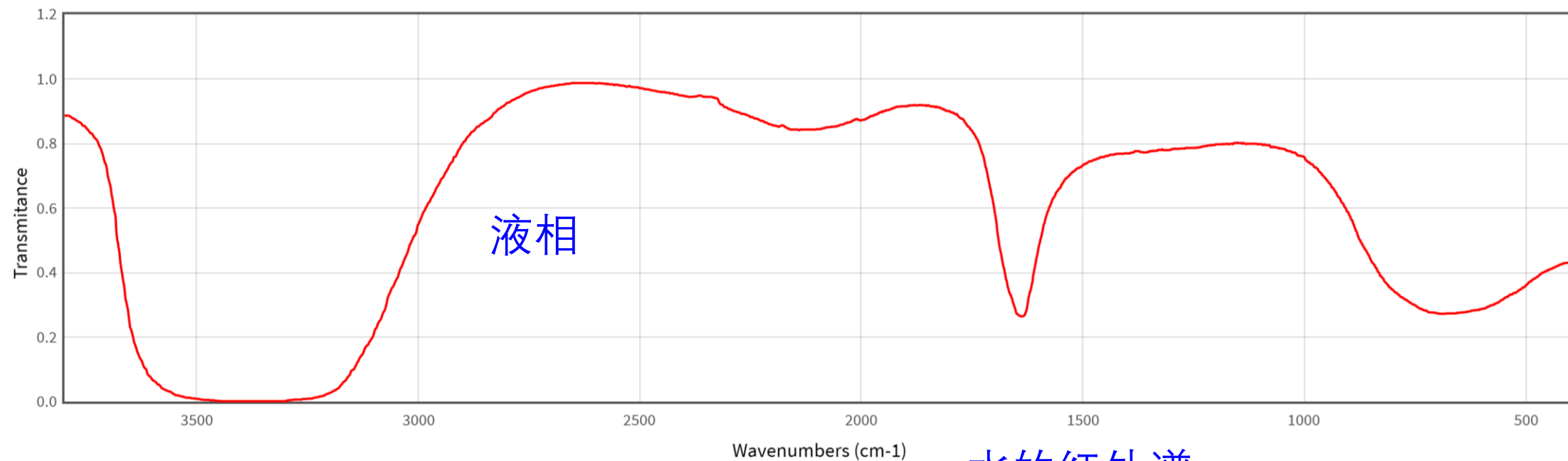
分子无法自由转动，振动-转动谱峰展宽，分辨率降低，抓取主要特征

- 溶液红外谱

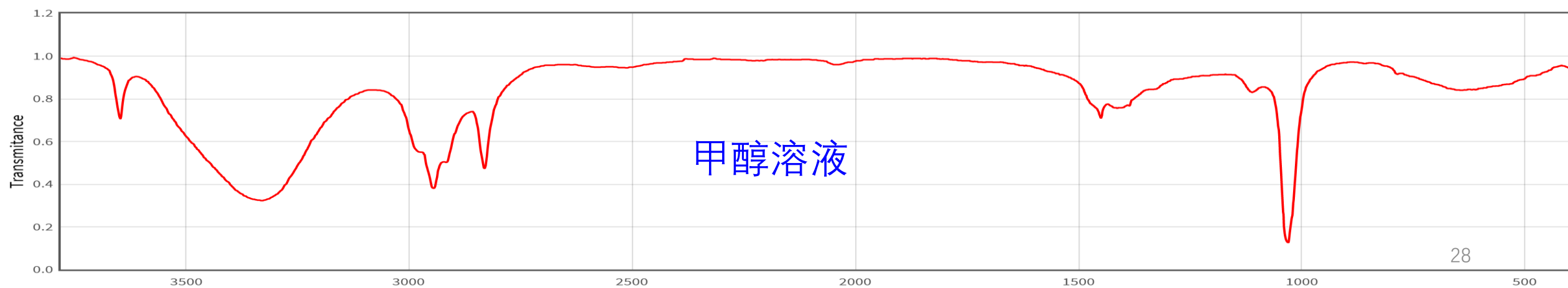
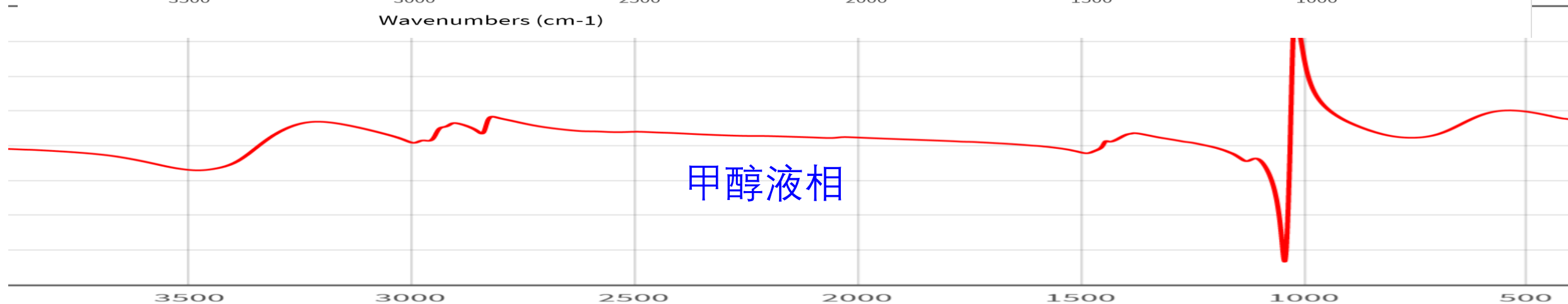
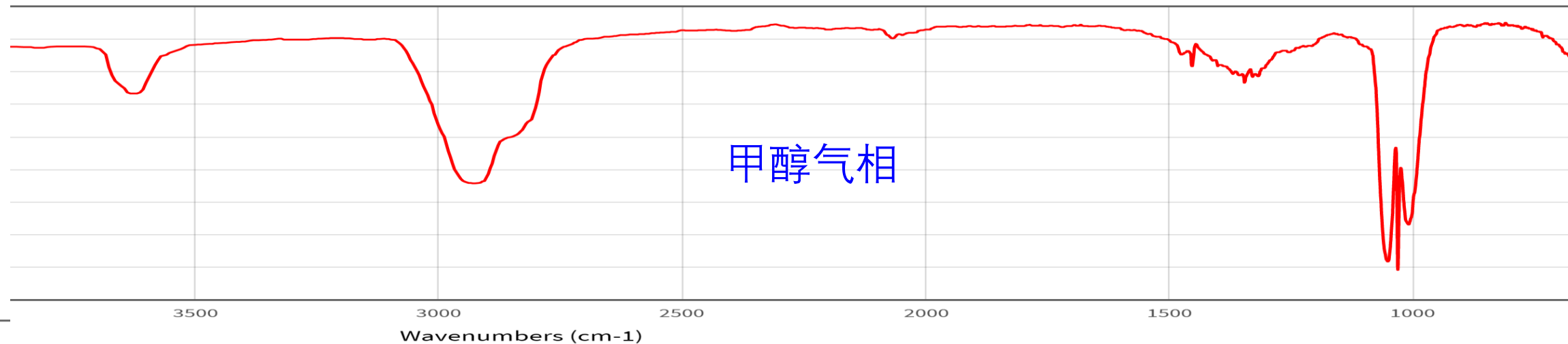
形成多个谱带，识别分子官能团的特征吸收谱带，通常在特定范围

液相水的红外谱





水的红外谱

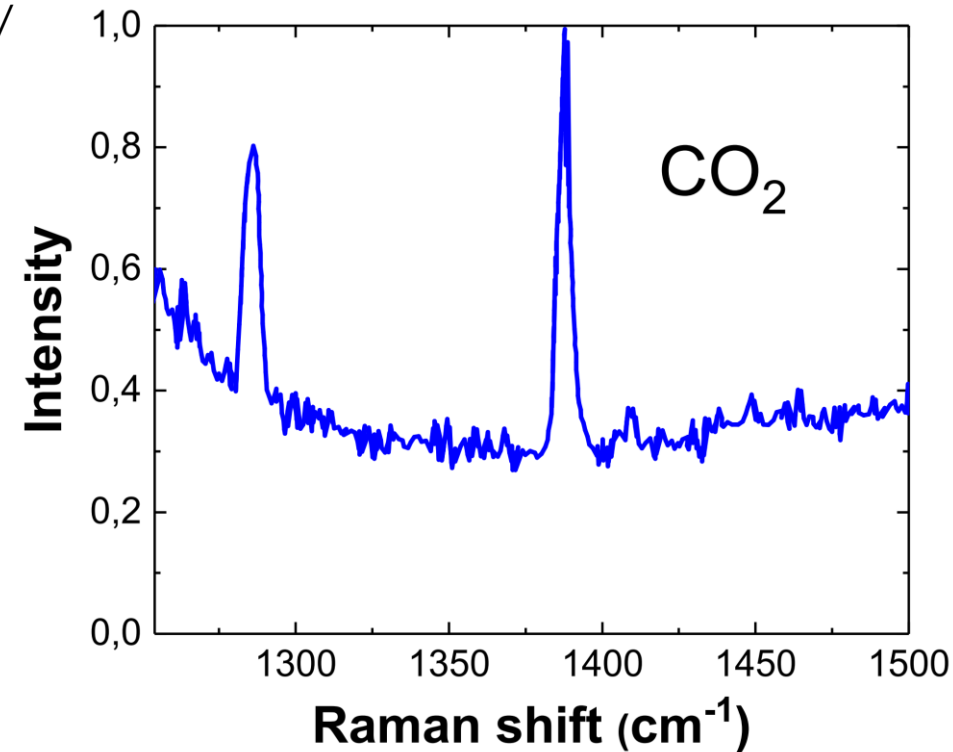


振动拉曼谱选律:

- 简正模式导致极化率发生改变
- $\Delta v_q = \pm 1$ (简正近似下)

有对称中心的分子, 红外、拉曼互斥原则:

某一简正模式不能同时有红外、拉曼活性



Infrared Spectrum

